Evaluating inertial road-surface classifiers for pavement screening: acquisition groups, model selection and performance estimation

Abstract

Inertial road-surface classifiers need validation at the scale of their intended deployment. We audit source structure and evaluate model selection and performance estimation as separate endpoints. The analyses use RoadSens, nine PVS acquisitions, two Mexico/Puebla route families and a date-held-out test of public Karlsruhe inertial data. RoadSens full-feature random-forest macro-F1 falls from 0.9945 under random-row validation to 0.4289 under recording holdout. PVS matched validation improves selected-model scenario performance but fails the declared two-axis estimation-error criterion. Its scenario annotation-interval recall rises from 34/60 to 52/60, accompanied by more non-bump alarms. On the untouched Karlsruhe date, buffered-date selection raises macro-F1 from 0.6782 to 0.7855 and reduces absolute estimation error from 0.2817 to 0.2072; both selected systems detect 7/12 annotated pothole intervals. These results distinguish candidate ranking, score estimation and event-screening performance, and support reporting them with explicit acquisition groups, training budgets and exposure denominators.

Keywords: pavement screening; inertial sensing; acquisition grouping; model selection; performance estimation

1. Introduction

Vehicle-mounted inertial sensors and smartphones can support frequent screening of road-surface anomalies between detailed pavement inspections. For an agency considering such a system, two questions arise: which classifier should be selected, and what performance should be expected when it is used under a different acquisition condition? A validation procedure answers both questions in many machine-learning studies, although candidate ranking and the accuracy of a reported score are distinct properties.

The engineering value of low-cost sensing depends on a clearly defined measurement target. Smartphone roughness studies have compared measurements with conventional instruments

(Sandamal and Pasindu 2022) and examined consistency across vehicle types, mountings and survey speeds (Yu et al. 2023). Road-event classification adds a further distinction between pavement distress, intentional surface geometry and ordinary travel. A pothole and a speed bump may both generate an inertial response, but they have different implications for maintenance. The present study evaluates classification of released road-event labels and the associated event-screening workload. The measurement targets are labelled surface events; roughness, distress severity and structural condition require their own reference measurements.

Acquisition structure matters because nearby sensor observations can share a road location, a vehicle response, an annotation interval and a recording configuration. Randomly withholding rows tests prediction with much of this information already represented in training. Withholding a complete acquisition instead tests a change in collection conditions. Spatial, temporal and hierarchical validation methods formalise these distinctions (Roberts et al. 2017, Ploton et al. 2020). Their use in pavement screening requires both an observable grouping unit and a statement of which deployment change that unit represents.

RoadSens-4M combines smartphone measurements, geographic and environmental information, and road-event annotations (Khandakar et al. 2026b, 2025b). It provides a useful setting for examining how label interpretation, feature representation and evaluation scale interact. The data descriptor reports a baseline on a selected subset; the present audit defines a complete, version-specific Combined v3 cohort, explicit label policies and saved training/test memberships. CKDP-PINNFed addresses GPS processing, privacy and federated road-anomaly modelling using RoadSens-related data (Khandakar et al. 2026a). The question here is how the available source structure supports the interpretation of classification scores.

Performance estimation is also an engineering concern in pavement modelling. Yao et al. (2022) combine predictive modelling with uncertainty analysis and interpretation for pavement performance evolution. In a road-event classifier, a validation score is an estimate of the performance of the model selected using that validation procedure. A procedure can choose a stronger model while producing a less accurate estimate of its score. Testing this distinction requires an outer domain that is excluded from candidate selection.

We examine this issue using a separate nine-acquisition PVS speed-bump task (Menegazzo and von Wangenheim n.d., 2023). The original PVS work already studied changes in vehicle, driver and environment. Here, random-window, whole-trip and domain-matched selectors choose among the same four candidates, and the selected models are assessed on complete outer domains. An additional Mexico/Puebla released-subset task (Aguilar-González and Medina Santiago 2025, 2026) provides a contrasting selection outcome on two documented route families. A further independent-source test uses public Karlsruhe vehicle-inertial data (Masino et al. 2018, Masino 2018), with a complete GPS-date group reserved for final testing and spatial buffering of repeated locations. Each dataset is analysed as a separate task.

The study links three levels of evidence: a RoadSens source and label audit; controlled comparisons of evaluation scale, context and temporal representation; and nested selection experiments that separate outer classification performance from validation-estimation error. The central question is whether a validation scheme that better matches a deployment grouping improves both endpoints. Annotation-interval detection and non-event alarm counts connect these score comparisons to screening workload. The resulting evidence informs how a pavement-screening study can report model choice, the reliability of its score and its event-level operating behaviour.

Figure 1 summarises the acquisition evidence, nested selection design and the distinction between classification performance and estimation accuracy.

2. Materials and methods

2.1. RoadSens population and labels

The audited Combined v3 snapshot contains 101 complete CSV sources: five normal-only files and 96 mixed/anomaly files, comprising 616,866 raw rows. The aggregate file `99.csv` is excluded, and the separate Isolated package is analysed outside this cohort. Retaining zero-based source rows 0, 10, 20 and so forth yields 61,730 candidate observations. Throughout this article, a *recording* denotes one source CSV. The release does not establish a one-to-one correspondence between a recording and an independently sampled road, trip, vehicle or device.

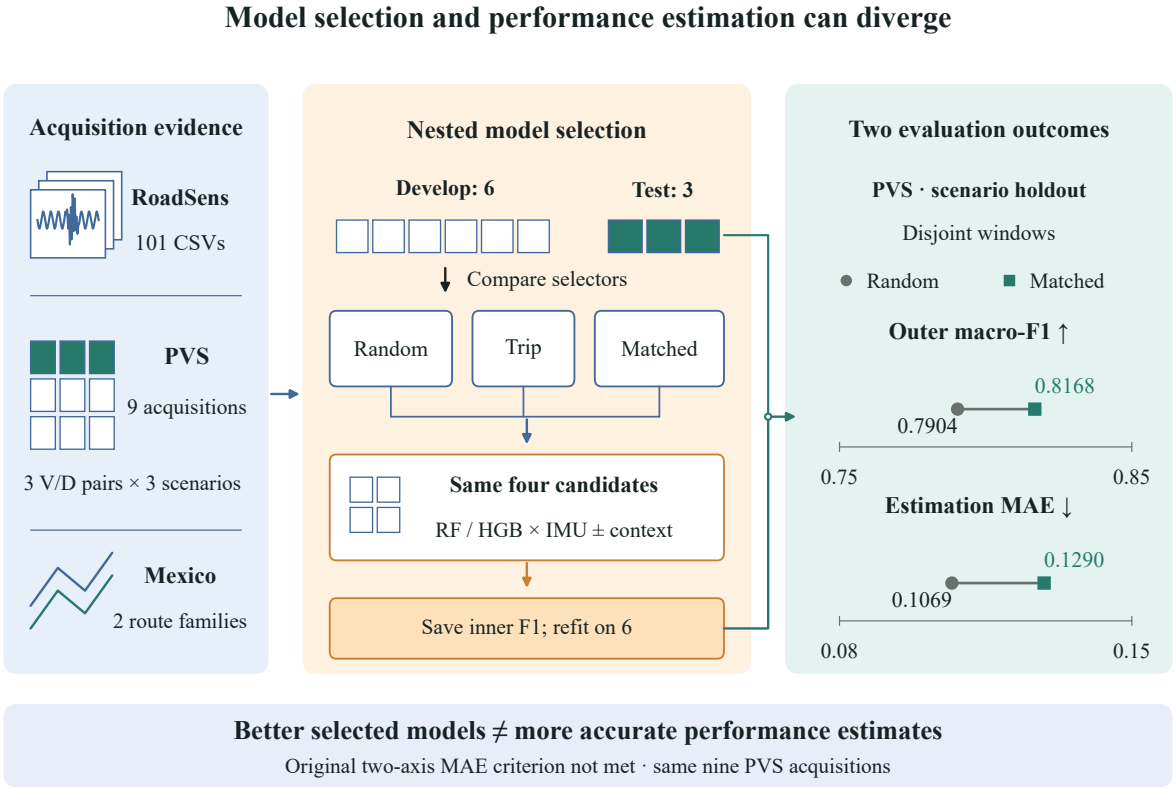


Figure 1: Graphical summary of the evaluation audit. RoadSens, PVS and Mexico/Puebla are separate tasks. PVS develops on six acquisitions and tests on three. The right panel gives three-domain scenario means with disjoint windows: matched selection increases outer F1 and estimation MAE. Matrix highlights, waveforms and route lines are schematic; V/D denotes vehicle/driver.

The primary policy retains normal labels from the normal-only directory and explicitly labelled pothole and bump observations from mixed recordings. The blank-as-normal sensitivity assigns blank mixed-record observations to Normal. Table 1 gives the resulting class populations. These policies represent different interpretations of the released annotations and use the same source observations. Source hashes, column identities and exact retained row indices are recorded with the analysis.

2.2. Representations and classifiers

The four main representations are full numeric (42 fields), context (eight geographic/environmental fields), instantaneous inertial measurement unit (IMU; 12 axes), and availability masks for the 42 numeric fields. Context comprises latitude, longitude, vertical accuracy, GIS elevation and four weather variables. Labels, timestamps, press duration and source identifiers are excluded from the predictors. Eight random-forest ablations examine GPS values, weather, elevation, numeric features without context, IMU with GPS or weather, GPS availability, and numeric values with availability masks.

For every fit, fields without a finite training value are removed and remaining non-finite values are replaced with training medians. A single zero-valued feature is used if no predictors remain. Availability-only features retain the parent implementation's underlying-field availability rule. Logistic regression additionally uses training-fitted scaling. Exact field ordering and the treatment of constant masks are specified in the supplementary methods.

The main classifiers are random forest (RF100), Extra Trees, histogram gradient boosting (HGB) and logistic regression. RF100 uses 100 trees, maximum depth 20, minimum leaf size 5, square-root feature sampling and balanced class weights. HGB uses 100 iterations, at most 31 leaves, learning rate 0.1, L2 regularisation 1 and balanced classes, with early stopping disabled. Additional forest capacity and training-budget configurations test sensitivity to the fixed settings. Complete parameters are in Supplementary Section S15.

2.3. RoadSens evaluation units

All main comparisons use five folds. Random-row evaluation uses stratified row folds. Recording-held-out evaluation keeps each CSV together and allocates one normal-only source to each test fold. Purged time-block evaluation divides each recording's full original time span into five equal-width bins before label filtering. A fold tests the corresponding bin in every recording and removes training observations in the same recording within an inclusive one-second interval of a test timestamp. Training can contain both earlier and later bins; this target is temporal separation within known recordings.

Methods within a policy, split and seed share the same eligible training rows, test rows and capped fit rows. The main fit cap is 20,000 rows, sampled in proportion to class frequencies using the frozen procedure. Seed 42 defines the principal comparison; seeds 43 and 44 provide additional random-row and recording-held-out checks. Saved source assignments and exact row memberships define the historical splits. These repeated analyses assess sensitivity within the same cohort.

The content audit links two sources when all 12 finite inertial axes match for at least 20 consecutive retained rows, after signed-zero normalisation. Labels and model scores are excluded from this rule. Three non-singleton components, {30,31}, {45,78} and {46,47,68,79}, produce 96 components overall. Component-held-out evaluation keeps connected sources together. A separate 100-file cohort removes only byte-identical source 47.

The explicit-anomaly control uses 28,478 labelled bump/pothole rows from the 96 mixed sources. Buffered spatial evaluation groups this population into globally anchored 100-m or 500-m cells after local coordinate projection. Five-fold stratified group assignments are followed by removal of training points within 100 m of any test coordinate, before the fit cap. Recording, content and spatial restrictions address different shared relations: spatial folds can contain different parts of the same recording or content component.

Two additional RoadSens diagnostics examine source-row strides of 5 and 20, and trailing one-second statistics computed from full-cadence inertial signals. The latter supplies 60 raw-axis statistics and uses identical fit/test anchors to the instantaneous-IMU control. Raw-row membership checks verify that training and test windows within a source are disjoint. A

same-runtime reconstruction of the original window implementation precedes the temporal comparison. Supplementary Sections S9 and S10 document these checks and their implementation history.

2.4. PVS acquisition groups and nested selection

PVS contains three vehicle/driver pairs crossed with three author-defined scenarios, yielding nine acquisitions (Menegazzo and von Wangenheim n.d., 2023). Vehicle and driver identities are inseparable in this design. We use the left-network dashboard accelerometer and gyroscope, sampled at nominally 100 Hz, together with the released sample-index labels. Structural checks verify equal sensor/label lengths, chronological timestamps and declared columns. The binary target combines asphalt and cobblestone speed-bump labels against the released `no_speed_bump` category.

Trailing 100-row windows have anchors 99, 124, 149 and so forth, with the target label taken at the final anchor. Mean, population standard deviation, RMS, range and maximum absolute magnitude for each of six axes yield 30 inertial features. The contextual variant adds anchor latitude, longitude and speed. The primary grid contains 43,204 windows, including 682 speed-bump anchors, from 1,080,905 source rows. A disjoint-window sensitivity uses anchors 99, 199, 299 and so forth, yielding 10,804 windows from the same acquisitions.

The four candidates combine RF100 or HGB with inertial-only or inertial-plus-context inputs. Hyperparameters follow the fixed settings above. Median imputation is fitted on the actual training rows; a deterministic class-proportional cap limits each fit to 20,000 windows. Test observations retain their original class distribution.

Each outer axis has three folds, holding out either one vehicle/driver pair or one scenario. Every fold uses six development acquisitions and three test acquisitions (Figure 2). Development data support three selectors: stratified random-window two-fold validation, leave-one-trip-out six-fold validation, and leave-one-remaining-domain-out two-fold validation matched to the outer axis. Candidates share exact split and cap indices. A selector chooses the maximum pooled inner out-of-fold (OOF) macro-F1, with a fixed candidate-order tie break. Choices are saved

before outer prediction. The selected candidate is refitted on the development population and evaluated on the held-out domain.

Let $\widehat{F}_{s,d}$ denote the saved inner estimate for selector s in outer domain d, and let $F_{s,d}^{\mathrm{out}}$ denote the outer macro-F1 of that same selected candidate. The estimation endpoint is

$$E_s = \frac{1}{3}\sum_{d=1}^{3}\left|\widehat{F}_{s,d} - F_{s,d}^{\mathrm{out}}\right|. \tag{1}$$

The classification endpoint is the mean of $F_{s,d}^{\mathrm{out}}$ across the three outer domains. The declared primary criterion requires matched validation to reduce E_s by at least 0.02 relative to random-window validation on *both* axes. This threshold is a practical decision rule. Signed bias is outer minus inner F1. Both axes and window grids reuse the same nine acquisition units.

2.5. PVS context and training-budget diagnostics

Two post hoc diagnostics were fixed after inspection of the initial PVS results. The first separates the additions of speed, latitude/longitude and their combination to the 30 inertial statistics. Speed is itself GPS-derived in this release; the coordinate block refers specifically to latitude and longitude. All parent outer splits, capped training memberships and model settings are retained.

The second uses the existing disjoint-window population and matches per-class training counts across all inner training pools and the outer training pool within each domain. The minimum available count of each class defines the common budget. One fixed subsample per split is shared by the four candidates; imputation uses that subsample. Validation and test observations are unchanged. The primary diagnostic estimates error against equally budgeted outer refits. A secondary target uses the historical full-development outer predictions for the newly selected candidate. These targets describe different refitted systems and are reported separately. Training composition changes together with budget.

2.6. Annotation-interval screening diagnostics

We additionally evaluate the frozen PVS selected systems as event-screening outputs. The nine source label sequences contain 60 contiguous speed-bump intervals, with 11, seven and two

per scenario for each vehicle/driver pair. These are labelled passages over features. A positive run ends at a non-positive row or a timestamp gap greater than 0.1 s. An interval is evaluable when it contains a saved outer-test decision anchor. Consecutive positive decisions on the 0.25-s grid form one alarm cluster; a negative decision or an excluded grid position separates clusters. Maximum one-to-one matching associates alarm clusters with annotated intervals using decision timestamps. The primary tolerance is zero; a 0.5-s expansion is a sensitivity check.

Alarm clusters outside every eligible interval are non-event alarms. Unmatched clusters associated with an interval are redundant notifications. Non-event rates use explicit non-bump exposure between the first and last assessed anchor, at the released 100-Hz cadence. Results pool counts across the three outer domains separately for each axis, covering every acquisition once per selector. Protocol definitions were fixed before computing these post hoc event outcomes. A subsequent reporting clarification separates redundant notifications from non-event alarms while retaining the initial total unmatched counts. Supplementary Section S16 supplies per-acquisition values and both tolerances.

2.7. Karlsruhe independent-source date holdout

The Karlsruhe release contains 53 IMU workbooks and 13 additional event excerpts (Masino 2018). The original acquisition system used an LSM9DS1 inertial sensor in a BMW 116d (Masino et al. 2018). The 53 IMU workbooks comprise 1,764,985 timestamped rows. Decoded GPS timestamps identify 15 workbooks in the 13 May 2016 group, 29 in the 20 June group and four in the 5 July group; five workbooks have no recoverable GPS date. The sensor's numeric epoch disagrees with the GPS calendar, so within-file numeric time defines durations and GPS time defines date groups. The excerpts and undated workbooks are excluded from this task. These file counts describe the release structure; repeated road features remain shared acquisition content.

The binary target uses the documented event codes 1 (good) and 3 (pothole). Other codes remain outside the labelled task, including code 0 (unknown) and unresolved code 7. Trailing one-second windows are evaluated every 0.25 s, using at least 50 raw rows, finite inertial axes and no timestamp gap above 0.1 s. The final source row at or before the decision provides its

label and context. Thirty inertial statistics and the optional coordinate/speed block follow the PVS feature definitions. This yields 5,226 eligible windows.

The May and June GPS-date groups supply development data. The July group is fixed as the final test before model fitting. A development window is removed when any raw coordinate in its span is invalid or lies within 100 m of an eligible final-test anchor. Random-window two-fold selection is compared with two-fold GPS-date selection, applying the same additional spatial buffer against each date-validation population. The four existing candidates and their settings remain fixed. Every inner fit uses the common minimum available per-class counts across all four training pools; exact subsamples and split memberships are saved before fitting. Validation populations retain their observed class distribution.

Each selector's pooled OOF macro-F1 determines its candidate, with the existing tie order. Saved choices precede final-test prediction. Primary outer refits use all spatially eligible development windows. A separate outer target retains the common inner training budget. We report the selected candidate's final macro-F1 and the absolute difference from its saved inner estimate for each target. The engineering diagnostic applies the same interval/alarm matching to retained good/pothole decisions. Its exposure denominator is explicit good-labelled raw time, integrating positive timestamp increments capped at 0.1 s. A single final GPS-date group supplies this independent-source test; Supplementary Section S17 gives source checks, memberships and all candidate outcomes.

2.8. Additional-source tasks

The Mexico/Puebla publisher supplement supplies four 9,000-position JSON arrays organised into two documented outbound/return route families (Aguilar-González and Medina Santiago 2025, 2026). Both directions of a route are held out together. Ten-position non-overlapping windows retain a unique majority label of at least six positions; any braking-containing window is excluded. This produces 2,504 non-event/pothole/bump windows. A 50-position proposal retained no target events and was replaced before fitting. The non-event class follows the release's blank-label convention.

Each window has 36 summaries of acceleration x/y/z and vector magnitude. RF100 and HGB are compared using random-window and whole-trip two-fold inner selection within the single development route, with matched per-class training counts. Both selectors share candidate-specific outer refits. Full-development and common-budget outer targets are evaluated separately. The source JSONs contain no timestamps, coordinates or vehicle/driver identifiers; route grouping follows the documentation. Supplementary Section S14 gives exact features, counts and source-version details.

A separate STRIDE v4 task summarises 17 bump/pothole recording types from one device (Khandakar et al. 2024, 2025a). It supplies a recording-type sensitivity in Supplementary Section S6. The RoadSens, PVS, STRIDE and Mexico tasks use their own annotations and fitted models.

2.9. Metrics, uncertainty and analysis provenance

RoadSens tables report arithmetic means of five fold-specific macro-F1 scores. Pooled OOF macro-F1 is computed separately from the combined held-out confusion matrix. For K fixed classes,

$$F_{\text{macro}} = \frac{1}{K}\sum_{k=1}^{K}\frac{2TP_k}{2TP_k + FP_k + FN_k}, \tag{2}$$

with zero assigned to an undefined class term. Balanced accuracy, class recalls and confusion counts accompany the primary metric in the supplement. The order of pooling and applying this nonlinear metric distinguishes pooled scores from fold means.

Conditional uncertainty for 26 RoadSens pooled contrasts uses 2,000 shared bootstrap draws of the 96 fixed content components, stratified into five normal-only and 91 mixed components, with seed 20261016. Confusion matrices are accumulated using each draw's multiplicities; the same draw weights compare both predictors. Percentile intervals describe sensitivity to this cohort's component composition while holding fitted models fixed. PVS summaries average three outer domains per axis, and Mexico summaries contain two route families; their results are descriptive at those grouping scales.

The RoadSens expansion followed exploratory screening. Initial PVS nested-selection and disjoint-window protocols were frozen before PVS fitting; context and common-budget diag-

nostics were defined after inspection of those outcomes. The Mexico protocol and exact index map were fixed before predictive fitting, following label-support checks. The Karlsruhe protocol, final date, buffer and candidate pool were fixed after source-quality inspection and before any Karlsruhe predictive fitting. Its inner training counts are matched across selectors. Protocol versions, preflight revisions, model reload checks and numeric reconstruction records are retained in the supplement. The analysis environment uses Python 3.13.12 and scikit-learn 1.9.0, with complete versions listed in Supplementary Section S1.

3. Results

3.1. Evaluation scale and representation in RoadSens

Full-numeric tree ensembles achieve random-row macro-F1 above 0.98 under the primary policy, compared with 0.4289–0.4569 under recording holdout (Table 2). RF100 changes from 0.9945 to 0.4289; its time-block score is 0.8095. The blank-as-normal sensitivity retains the same ordering for the tree ensembles, at lower score levels. Logistic regression has lower random-row performance and a smaller evaluation contrast. Figure 3 summarises the model and split comparisons.

Same-runtime RF random-minus-recording fold-mean differences remain positive across seeds 42–44: 0.5656, 0.5387 and 0.5354 under the primary policy, and 0.5360, 0.5548 and 0.5148 under blank-as-normal. For pooled OOF predictions, the primary RF difference is 0.5827, with a conditional 95% component-resampling interval of [0.4363, 0.6960]. Its blank-as-normal counterpart is 0.5455 [0.4881, 0.6042]. These intervals concern pooled contrasts; Table 2 reports fold means.

Source-field availability is associated with source type: all five normal-only sources lack complete GPS, whereas the 96 mixed sources have finite GPS. The four weather fields are constant within each recording. Availability-only RF yields primary random/recording scores of 0.6095/0.6019, compared with 0.2638/0.2720 under blank-as-normal. The added Normal observations in the latter policy change the relationship between class and source availability.

Context effects vary with label policy and representation (Table 3). GPS values yield recording-held-out macro-F1 of 0.6984 under the primary policy and 0.3508 under blank-as-normal. Numeric values with availability masks yield 0.6797 and 0.3885. Removing the eight context fields reduces RF recording-held-out scores in both policies. The paired pooled 34-field-minus-full contrasts are -0.1179 [-0.2486, -0.0208] and -0.0610 [-0.1064, -0.0185], respectively. Context-only versus full-feature contrasts also differ across classifiers, as detailed in Supplementary Section S4.

The explicit bump/pothole task retains a large contrast after excluding normal-only sources: full-numeric RF scores 0.9948 for random rows and 0.5189 for recording holdout (Table 4). Availability-only scores are 0.4143 and 0.4029. The normal-directory distinction therefore accounts for only part of the structure associated with the full-feature evaluation contrast.

One-second inertial statistics improve all eight matched temporal comparisons (Table 5). Recording-held-out fold-mean improvements range from 0.0631 to 0.0794; the resulting scores are 0.4023–0.4173. For example, primary RF increases from 0.3410 with instantaneous IMU axes to 0.4072 with temporal statistics. Time-block RF increases from 0.4454 to 0.5929. The representation comparison shows useful temporal information in the inertial signals under verified within-source raw-input separation.

3.2. Source, content and sampling controls

The source audit identifies byte-identical files 46 and 47 and eight finite-inertial matching edges in the three non-singleton content components. Longest retained-row matches include 427 rows for {30,31}, 90 for {45,78} and 246 within {46,47,68,79}. Component regrouping produces six positive and six negative pooled changes across 12 matched-cohort comparisons, approximately -0.0462 to $+0.0340$. Removing source 47 leaves the large full-feature random/recording contrast in the 100-file cohort. Shared content is present, while regrouping also changes training and fold composition.

Forest capacity, fit caps and sampling density give further checks (Supplementary Section S15). Uncapped RF100 recording scores are 0.4307/0.3880 for the primary/blank-as-normal policies, close to the capped 0.4289/0.3954 references. RF300 capped scores are 0.4774/0.3847.

Source-row strides of 5, 10 and 20 give primary full-numeric recording scores of 0.4386, 0.4289 and 0.4086; corresponding instantaneous-IMU scores are 0.3379, 0.3410 and 0.3386.

The buffered spatial task yields 350 cells at 100 m and 119 at 500 m, with observed training points more than 100 m from test coordinates. Full-numeric RF scores are 0.6349/0.6215 for the two grids, and context-only RF scores are 0.6450/0.6702. These values reflect a geographic separation target that retains other acquisition relationships. The temporal, capacity, content and spatial controls together show that the evaluation contrast depends on several aspects of the predictive system and its grouping structure.

3.3. PVS: selected-model performance and estimation error

In the primary PVS grid, random-window validation exceeds outer scores on average for all eight candidate/axis combinations, with mean differences of 0.0443–0.1696. Candidate-level means are in Supplementary Section S15. Table 6 reports each selector's estimate, selected-model outer F1, estimation MAE and signed bias.

For scenario holdout, matched selection increases mean outer macro-F1 from 0.7640 to 0.8359, while estimation MAE decreases from 0.1696 to 0.1357. For vehicle/driver holdout, outer F1 remains 0.8679 and MAE decreases from 0.0530 to 0.0445. The MAE reductions are 0.0339 and 0.0085 after rounding. The declared requirement of at least 0.02 on both axes is therefore not met. Whole-trip validation yields scenario MAE of 0.1390, compared with 0.1357 for matched validation, and vehicle/driver MAE of 0.0304, compared with 0.0445.

On disjoint windows, matched selection improves outer F1 from 0.7904 to 0.8168 for scenario holdout and from 0.8380 to 0.8522 for vehicle/driver holdout. Estimation MAE instead rises from 0.1069 to 0.1290 and from 0.0489 to 0.0535, respectively. Figure 4 separates these classification and estimation outcomes. The two grids differ in window counts and training populations, while using the same acquisitions.

The context-block diagnostic also varies across domains and classifiers. In primary vehicle/driver holdout, the combined context block gives the highest means for RF and HGB, 0.8352 and 0.8679. In scenario holdout, its coordinate-given-speed contrast is -0.0898 for RF and -0.0716 for HGB. On disjoint windows, RF's scenario coordinate-given-speed contrast is

+0.0147. Full block and domain results are retained in Supplementary Sections S12 and S15. The observed sign changes concern retrained feature representations.

3.4. PVS annotation-interval detection and alarm burden

All 60 positive annotation intervals contain an assessed decision anchor. Under vehicle/driver holdout, the three selectors choose systems that detect 59/60 intervals, with 71 non-bump alarm clusters over 2.9522 non-bump hours (24.05 per hour) and 14 redundant event-associated clusters. Under scenario holdout, random-window and whole-trip selection detect 34/60 intervals, with 22 non-bump clusters (7.45 per hour) and 14 redundant clusters. Matched selection detects 52/60 intervals, with 51 non-bump clusters (17.28 per hour) and 18 redundant clusters (Table 9). Interval recall therefore rises from 56.7% to 86.7% alongside a higher non-bump alarm burden. The 0.5-s matching sensitivity preserves all pooled hit counts. It reclassifies five vehicle/driver alarms and one scenario alarm per selector as event-associated, preserving the scenario ranking in non-event burden. These diagnostics reuse the existing nine acquisitions and saved hard decisions.

3.5. Common-budget diagnostic

The common-budget comparison uses 2,917–3,763 non-bump windows and 18–57 bump windows per fit, depending on outer domain. Table 7 keeps the two outer targets separate. With equally budgeted outer refits, random-to-matched MAE increases from 0.0140 to 0.0267 for vehicle/driver holdout and from 0.1627 to 0.1724 for scenario holdout. Outer F1 changes from 0.8468 to 0.8526 and from 0.7083 to 0.6951, respectively.

For the historical full-development outer target, random-to-matched MAE changes from 0.0400 to 0.0390 for vehicle/driver holdout and from 0.0941 to 0.1462 for scenario holdout. Matching counts therefore changes the error levels and selected systems, but does not produce a consistent matched-selector advantage. The diagnostic retains sparse positive-class training support and the existing acquisition domains.

3.6. Mexico/Puebla route-family task

Both selectors choose RF in both held-out route families. Selected-model outer macro-F1 is 0.9934 for the urban family and 0.9960 for the suburban family (Table 8); the outer selection contrast is exactly zero. Estimation MAE is 0.0019 for random selection and 0.0011 for whole-trip selection with full-development refits. The common-budget sensitivity yields 0.0015 and 0.0007.

This released subset thus has near-ceiling classification scores and small estimation errors under both selectors. Its outcome contrasts with PVS, where matched selection changes the selected classifier in some domains. The two route families, short annotation runs and documentation-derived grouping define the scale at which the Mexico result can be interpreted.

3.7. KIT: a final date with buffered spatial separation

The final test contains 917 windows (835 good and 82 pothole) from the four July workbooks. Spatial purging removes 20 of the 4,309 development windows, leaving 4,123 good and 166 pothole windows. The closest retained full training window is 922.43 m from a test anchor; both date-validation separations exceed 2,432 m. Every inner candidate fit uses 1,613 good and 17 pothole windows.

Random-window selection chooses HGB with context, with inner macro-F1 0.9598. GPS-date selection chooses inertial-only HGB, with inner macro-F1 0.5783. Full-development final scores are 0.6782 and 0.7855, giving absolute estimation errors of 0.2817 and 0.2072, respectively (Table 10). With common-budget outer refits, the corresponding final scores are 0.6699 and 0.7595, and errors are 0.2899 and 0.1812. Thus, the date selector chooses a stronger system and produces a closer estimate on this final date under both refit targets.

The full-development systems classify 20/82 and 35/82 pothole anchors correctly, respectively, with no good anchors classified as potholes. Both detect seven of the 12 evaluable pothole annotation intervals. All five intervals in workbook 50 are missed; the seven detected passages occur in workbooks 51 and 52, which share a geographic neighbourhood. The date-selected system adds one redundant event-associated cluster (Table 9). Both have zero observed non-event clusters during 3.47 minutes of assessed good-labelled exposure. That exposure is concentrated

in one good-labelled workbook, and unknown-labelled travel is outside the task. The observed zero therefore describes this finite labelled exposure. The 0.5-s sensitivity gives the same interval hits and alarm categorisation. The additional correct decisions occur within intervals that both systems detect.

4. Discussion

4.1. Implications for pavement-screening evaluation

The RoadSens results show that a high score on held-out observations within known recordings can coexist with much lower performance on withheld sources. This matters when a pavement agency uses a benchmark to choose a screening model for another survey. A random-row result describes interpolation under the acquisition relations shared by its split. A recording result describes transfer across the available source files. A spatial result describes separation between observed coordinate neighbourhoods. Reporting these targets explicitly makes the score useful for the corresponding engineering decision.

The choice of target should follow the intended service. Reinspection of a known network can use geographic context as part of its design; deployment on new routes or collection platforms requires metadata that identify those changes. Location, speed and signal availability should therefore be evaluated against the declared target. Here, removing contextual fields decreases some held-out scores, and coordinate/speed additions have mixed signs across PVS domains. Those findings favour task-specific input assessment over a single feature-removal rule.

Label definitions are equally consequential. Blank-as-normal changes both the class population and its relationship with source availability. Separating an unlabelled interval from a verified normal observation is necessary when interpreting false alarms and normal-class performance. Likewise, speed bumps are intentional road features. Distinguishing them from potholes is relevant to screening, while severity and maintenance priority require additional reference information. The event diagnostics add a further distinction: PVS scenario recall improves at a higher alarm cost, whereas the Karlsruhe window-score improvement leaves the number of de-

tected intervals unchanged. A screening comparison should therefore pair classification scores with missed intervals, non-event alarms, redundant notifications and their assessed exposure.

4.2. Selecting a model and reporting its expected score

The PVS case study separates two decisions that are often combined. Matched validation selects systems with higher scenario-held-out F1 under both window grids. Its saved validation score is nevertheless less accurate than the random-window estimate on both disjoint-window axes. A stronger selected classifier and a more accurate performance estimate therefore require separate evaluation endpoints.

The common-budget diagnostic shows that training-count alignment affects the comparison without resolving the estimation pattern. Whole-trip and matched validation also have different training partitions and class support. Evaluation should report the training population associated with an estimate and the refit population associated with the deployed model. The Mexico task adds an informative alternative: both selectors choose the same RF and estimate its high outer score closely. The independent Karlsruhe date test favours the date selector for both selected-model score and estimation error, including under matched inner/outer training counts. Its unchanged interval hit count adds a third outcome alongside model ranking and score estimation. Selector differences are thus contingent on candidate rankings, the available domains and the engineering endpoint.

For pavement-screening reports, the practical output is a pair of quantities: the outer-domain score of the selected system and the discrepancy between that score and its saved validation estimate. The first supports candidate choice; the second describes how accurately the study can communicate expected performance. Keeping them separate also makes training-budget and window sensitivities easier to interpret.

4.3. Evidence scope and further field evaluation

The RoadSens population is one package version with five normal-only sources. Its CSV and content groups are observable source units, with incomplete road, vehicle and device provenance. Conditional bootstrap intervals retain the fitted predictors and this finite component structure.

PVS supplies nine acquisitions, three outer domains per axis and perfectly linked vehicle/driver identities. Mexico supplies two documented route families whose relationship to the larger published collection remains unresolved. Karlsruhe adds a separately collected source, three recoverable GPS-date groups and one final test date after spatial purging. Its single vehicle and sparse, selectively labelled test exposure define the transfer target. These grouping counts set the scale of the empirical evidence; windows, fitted models and repeated annotation intervals supply dependent observations within it.

The analyses use released annotations, conventional fixed candidates and hand-specified representations. The temporal and context diagnostics change the fitted predictive systems; they quantify representation sensitivity. The post hoc PVS diagnostics use inspected outcomes and serve to locate sources of variation. The study establishes how evaluation choices affect these benchmark tasks, with field generalisation requiring additional acquisition evidence.

The next field stage should cross vehicle and driver identities, retain dates and route/event identifiers, and separate repeated passes over the same road feature. A development cohort can define a revised selection or estimation rule, followed by a frozen evaluation on untouched whole acquisitions. Planning acquisition numbers against a desired uncertainty range would make that stage informative for the intended survey conditions. The present interval-based diagnostics quantify labelled-event coverage and notification workload. A field service additionally requires independently adjudicated physical events, complete normal-travel exposure and reliable travelled-distance records to assess maintenance inspection workload.

5. Conclusions

RoadSens recording holdout gives substantially lower full-feature classification scores than random-row evaluation, while temporal inertial summaries improve held-out performance. Context, content and sampling controls produce varied changes, showing the importance of specifying labels, source relationships and model inputs together. In PVS, domain-matched validation improves selected-model scenario performance but does not meet the declared two-axis estimation-error criterion; estimation MAE increases on both disjoint-window axes and under the equal-budget outer diagnostic. The Mexico/Puebla subset gives identical selected models

and small errors for both selectors. The independent Karlsruhe final date favours buffered-date selection for classification and estimation, while both selected systems detect seven of 12 pothole intervals. PVS scenario detection improves alongside a higher non-bump alarm rate. For sensor-based pavement screening, these results support reporting model-selection performance and performance-estimation accuracy separately, with the acquisition grouping and refit budget attached to each reported score.

Data availability statement

Source data are available from the cited RoadSens, PVS, STRIDE, Mexico/Puebla and Karlsruhe providers. The Karlsruhe archive is identified by DOI 10.5281/zenodo.1461243 and SHA-256 `f7f7cc6b017977d2019f1eaaf9e90d6ad6132506965bafba03c12e6b053845d3`. Review materials include protocols, compact results and source hashes; the public code repository URL is withheld in this version to support anonymous review. Third-party raw signals, coordinates, complete prediction arrays and fitted models are excluded from the public distribution. Reanalysis requires the separately obtained source data and hash-matched inputs described in the repository documentation.

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Disclosure statement

The authors report no competing interests.

Ethical and data-use considerations

This study analyses previously released road-sensor datasets. No new participants were recruited and no interventions were performed. Data were obtained from the cited providers under their

source terms. The public research outputs report aggregate results and coded acquisition groups; raw location traces and third-party sensor records are excluded from redistribution.

Use of generative artificial intelligence

OpenAI Codex (GPT-6-family models, including GPT-6 Luna; accessed October–November 2026) was used to assist implementation and English-language presentation through literature searches, analysis and plotting code, manuscript drafting, language editing and LaTeX preparation.

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Table 1: RoadSens supervised populations under the two label policies.

Policy	Normal	Pothole	Bump	Total supervised rows
Primary	4,407	11,346	17,132	32,885
Blank-as-normal sensitivity	33,252	11,346	17,132	61,730

Table 2: Full-numeric macro-F1: five-fold arithmetic means, seed 42.

Policy	Model	Random rows	Purged time blocks	Heldout recordings
Primary	RF100	0.9945	0.8095	0.4289
Primary	Extra Trees	0.9835	0.7802	0.4299
Primary	HistGB	0.9981	0.8460	0.4569
Primary	Logistic regression	0.7519	0.7042	0.4324
Blank-as-normal	RF100	0.9314	0.5775	0.3954
Blank-as-normal	Extra Trees	0.8715	0.5340	0.3978
Blank-as-normal	HistGB	0.9350	0.6198	0.4150
Blank-as-normal	Logistic regression	0.4564	0.4109	0.3404

Table 3: RF feature ablations: five-fold mean macro-F1, seed 42. Sensitivity denotes the blank-as-normal policy.

Feature group	Primary random	Primary temporal	Primary recording	Sensitivity random	Sensitivity temporal	Sensitivity recording
GPS values	0.9985	0.8672	0.6984	0.9546	0.6180	0.3508
Weather	0.5465	0.5138	0.2958	0.4082	0.3586	0.2279
GIS elevation	0.6188	0.4354	0.3037	0.5533	0.4217	0.3888
Numeric minus context	0.9362	0.6739	0.2905	0.7792	0.4431	0.3362
IMU + GPS	0.9740	0.7953	0.7113	0.8398	0.5404	0.3929
IMU + weather	0.7814	0.6120	0.3059	0.5993	0.4155	0.3149
GPS availability	0.5233	0.5566	0.5431	0.2228	0.2310	0.2315
Numeric + availability	0.9926	0.8003	0.6797	0.9175	0.5683	0.3885

Table 4: Explicit bump/pothole classification: RF100 five-fold mean macro-F1.

Features	Random	Temporal	Recording
Full numeric	0.9948	0.7271	0.5189
Context	0.9991	0.8153	0.5559
Availability only	0.4143	0.4089	0.4029
Instantaneous IMU	0.6321	0.5107	0.5240

Table 5: Temporal-context comparison: five-fold mean macro-F1 with identical fit/test anchors. The one-second representation uses current and past input.

Policy	Split	Model	Instantaneous IMU12	One-second statistics60
Primary	Recording	RF100	0.3410	0.4072
Primary	Recording	HistGB	0.3379	0.4173
Primary	Time-block	RF100	0.4454	0.5929
Primary	Time-block	HistGB	0.4419	0.6015
Blank-as-normal	Recording	RF100	0.3448	0.4100
Blank-as-normal	Recording	HistGB	0.3392	0.4023
Blank-as-normal	Time-block	RF100	0.3591	0.4256
Blank-as-normal	Time-block	HistGB	0.3581	0.4294

Table 6: Primary PVS selector outcomes: three-domain means per axis. MAE compares each selected candidate's saved inner estimate with its own outer score; signed bias is outer minus inner F1.

Axis	Selector	Mean estimate	Mean outer F1	MAE	Outer-minus-inner bias
Scenario	Trip	0.8795	0.7640	0.1390	-0.1155
Scenario	Matched	0.7928	0.8359	0.1357	0.0431
Scenario	Random	0.9336	0.7640	0.1696	-0.1696
Vehicle	Trip	0.8598	0.8679	0.0304	0.0081
Vehicle	Matched	0.8318	0.8679	0.0445	0.0361
Vehicle	Random	0.9210	0.8679	0.0530	-0.0530

Table 7: Post hoc PVS common-budget diagnostic: three-domain means. The common-budget and historical-full-budget outer targets describe different refitted systems.

Axis	Selector	Inner F1	Outer F1	MAE
Common budget outer target				
Vehicle	Random-window	0.8581	0.8468	0.0140
Vehicle	Leave-one-trip	0.8402	0.8468	0.0217
Vehicle	Matched	0.8289	0.8526	0.0267
Scenario	Random-window	0.8699	0.7083	0.1627
Scenario	Leave-one-trip	0.8389	0.7141	0.1428
Scenario	Matched	0.7740	0.6951	0.1724
Historical full budget outer target				
Vehicle	Random-window	0.8581	0.8485	0.0400
Vehicle	Leave-one-trip	0.8402	0.8485	0.0477
Vehicle	Matched	0.8289	0.8522	0.0390
Scenario	Random-window	0.8699	0.7986	0.0941
Scenario	Leave-one-trip	0.8389	0.8064	0.0768
Scenario	Matched	0.7740	0.8168	0.1462

Table 8: Mexico/Puebla full-development refit outcomes by documented held-out route. Both selectors share the same selected outer model.

Held-out route	Inner selector	Chosen	Inner F1	Outer F1	Abs. error
Urban (1+2)	Random	RF	0.9922	0.9934	0.0012
Urban (1+2)	Whole-trip	RF	0.9922	0.9934	0.0012
Suburban (3+4)	Random	RF	0.9934	0.9960	0.0026
Suburban (3+4)	Whole-trip	RF	0.9950	0.9960	0.0010

Table 9: Annotation-interval coverage and alarm burden of selected full-development systems. Zero matching tolerance.

Selector	Hits / intervals	Non-event clusters	Clusters per negative h	Redundant clusters	Negative exposure (h)
PVS vehicle/driver					
Random-window	59/60	71	24.05	14	2.9522
Matched	59/60	71	24.05	14	2.9522
PVS scenario					
Random-window	34/60	22	7.45	14	2.9522
Matched	52/60	51	17.28	18	2.9522
Karlsruhe final GPS-date group					
Random-window	7/12	0	0.00	0	0.0579
Buffered-date	7/12	0	0.00	1	0.0579

Higher interval coverage and lower alarm rates are desirable. Negative exposure means explicit non-bump time in PVS and explicit good-labelled time in Karlsruhe. PVS whole-trip results equal random-window results. Each PVS axis reuses the same nine acquisitions; Karlsruhe has one final date and only 3.47 minutes of assessed good exposure. Intervals are labelled passages, with repeated physical features possible.

Table 10: Karlsruhe selection and estimation on the final GPS-date group.

Selector	Selected candidate	Inner F1	Final F1	Absolute error
Full-development outer refit				
Random-window	HGB + context	0.9598	0.6782	0.2817
Buffered-date	HGB inertial	0.5783	0.7855	0.2072
Common-budget outer refit				
Random-window	HGB + context	0.9598	0.6699	0.2899
Buffered-date	HGB inertial	0.5783	0.7595	0.1812

The final group is 5 July 2016, decoded from GPS time. Higher final macro-F1 and lower absolute estimation error are desirable. Full refits use 4,289 development windows; common-budget refits use 1,613 good and 17 pothole windows. Both selectors use these same per-class counts for each inner fit. A single held-out date supplies each final score.

PVS acquisition grouping and nested evaluation

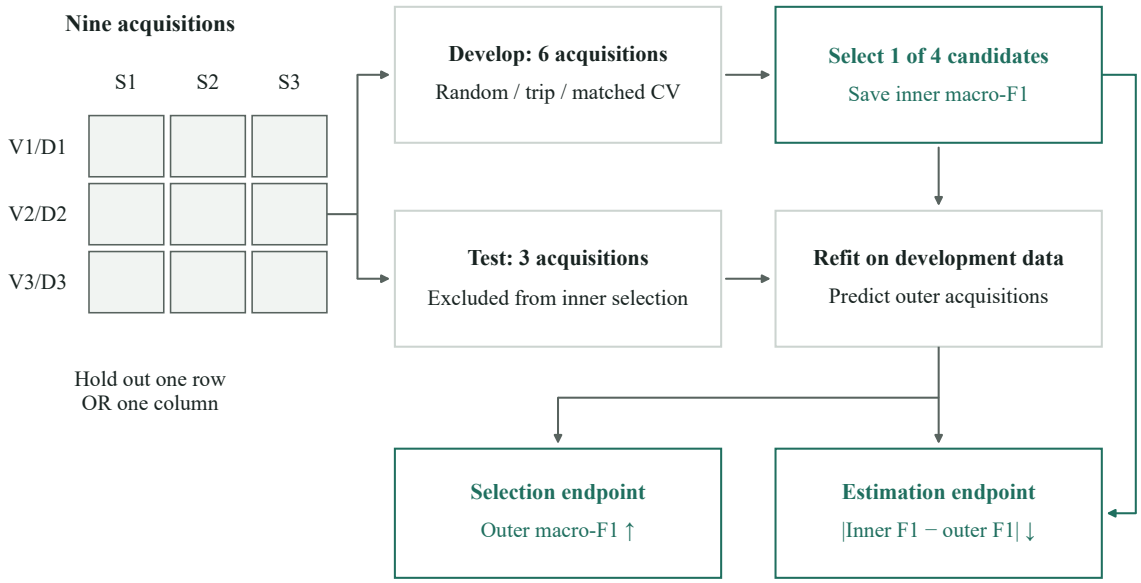


Figure 2: PVS acquisition grouping and nested evaluation. Each cell is one acquisition; S denotes scenario and V/D a vehicle/driver pair. A vehicle/driver-held-out fold excludes a row and a scenario-held-out fold a column. Each selector chooses among four candidates by pooled inner OOF macro-F1 on six development acquisitions. The saved estimate and the same candidate's outer score define an absolute error, averaged across three domains per axis. Refitting uses development data only. Vehicle and driver effects are linked, and both axes reuse nine acquisitions.

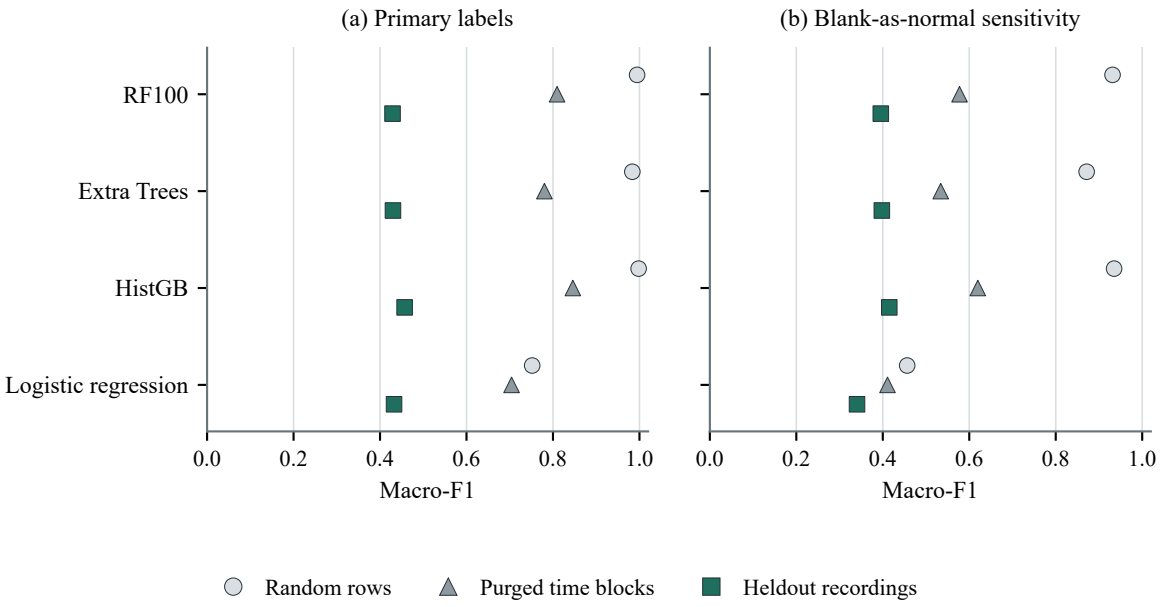


Figure 3: RoadSens full-numeric five-fold mean macro-F1 across four classifiers and three evaluation scales. Symbols denote point estimates; the same sources contribute to the three split designs. Exact values are in Table 2.

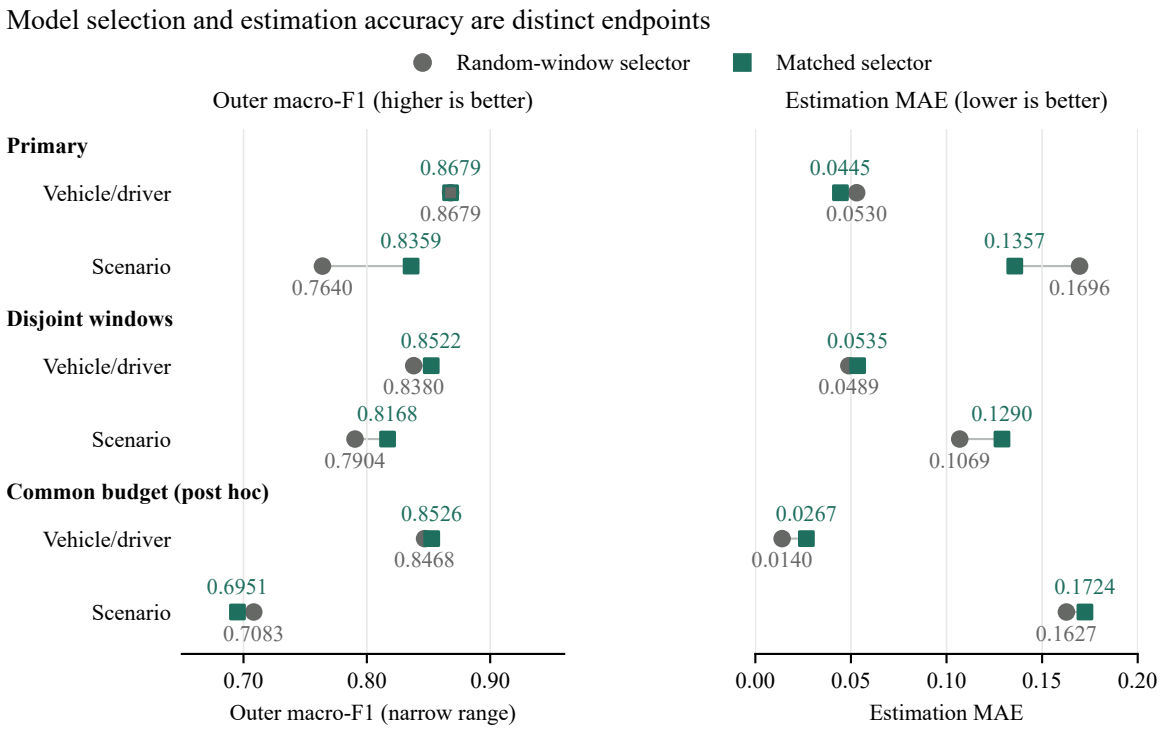


Figure 4: PVS selected-model classification performance and estimation error. Grey circles denote random-window selection and teal squares matched-domain selection. Segments connect selector summaries within the same analysis and held-out axis. Values are three-domain means from the same nine acquisitions. Primary and disjoint-window rows use full-development outer refits; the post hoc common-budget rows use equally budgeted outer refits. The left axis is restricted to 0.65–0.96 and the right starts at zero. The declared MAE-reduction requirement of at least 0.02 on both axes is unmet.